

Indigene: Integration and Discovery in Gene Networks

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Abstract

Functional genomic data have created a need for tools to integrate and analyze gene networks within a single framework. We focus on the model *Drosophila melanogaster* from a systems biology perspective, using a logic-theoretic (extensional) framework to combine disparate data sources with the goal of discovering gene function. In this framework, we define genes as externally annotated units (from FlyBase), and their interactions are defined as functional connections in a common genetic pathway, or which participate in a common process. Multiple sources of data such as microarray coexpression, protein-protein interaction, genetic interaction, transcription factor binding site overlap, and RNAi phenotype, are combined. These overlaps both support inferences from individual data sets, and further refines groups of genes into coherent clusters. For this integration, we use simple deductive rules based on expert knowledge of the datasets used, and allow a wide latitude for the user to set thresholds of belief. The system is also readily extensible to new sources of data. Using these techniques, we find that new functional connections can be made between genes because of the multiple types of information used. Using GO annotation, we find significant enrichment for clusters of functionally related genes, demonstrating that the approach taken has successfully integrated disparate genomic-scale data. An illustration is given imputing a tentative biological process for two proteins with no known function.

1 Introduction and Motivation

Recent results in combining high-throughput gene data indicate that the approach is biologically worthwhile [20, 16, 2, 23]. This *systems biology* perspective—bringing together multiple, disparate, and broad interactions—can help us fill-in some of the details of biological phenomena, for example, gene function. Typically, large, complex networks are built where nodes denote some biological entity (like genes) and edges are multiple kinds of relationships between them as indicated by, say, microarray assay and protein/protein. This approach is so powerful that it even allows us to make sense of local phenomena. For instance, we might be able to refine our understanding of the roles some individual genes play.

Systems biology can also aid in an experimental setting. As hypotheses are tested, the networks in turn can be re-tuned to test further hypotheses and so on. Another advantage is that it can be used as a discovery tool. Biologist can “roam” among the data, able to survey many times more landscapes than would be possible in flat systems. Often times, they make connections with other kinds of data outside of the network that they themselves are familiar with—perhaps to begin elucidating the function of unknown genes.

In practice these systems combine microarray assay and protein-protein interactions from a single organism, usually *C. elegans* or *S. cerevisiae*. The network is often further annotated with usually one more dimension, often RNAi phenotypic profiles or GO terms. Because of the size and nature of these networks, machine learning techniques are employed to help draw inferences. We believe reflecting on this choice, however briefly, is important because it sets explicit constraints on not only the data, but also on the inference procedure as well. While no uniform choices exist, two different kinds of approaches can be distinguished [18]: extensional and intensional. Roughly speaking, extensional means data are cast into a logic-theoretic framework. If two data A, B are to be combined, then some kind of operator like AND is used to produce a weighted combination, *e.g.* $A \text{ AND } B$. Intensional,

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on the other hand, means data are cast into a set-theoretic framework. The meaning of combining A, B becomes, say, a joint probability $P(A, B)$ by viewing A, B as probabilistic events and their combination as an intersection. Both approaches have advantages and disadvantages, and both have been used with success in systems biology, though there seems to be a predominant use of intensional. Intensional systems deal with uncertainty, for instance, rather easily. Extensional systems are appealing, though, because generally they require biologists to make few explicit assumptions. Furthermore, a number of inference procedures are straight forward to understand and are computationally tractable in the formal sense.

Systems biology still faces a number of significant challenges, and we list among them only a select few. First, no clear, formal consensus has arisen on how to combine and then make inferences over the data. Second, the networks are static and, therefore, can only provide a snapshot of some phenomena. Third, data often are noisy, sparse and incomplete.

In this ongoing project we describe some initial success from our integrated gene network (Indigene). We will discuss our data, methods, integration philosophy in building Indigene, and report some potentially interesting findings on two previously unannotated genes. We believe this work is significant for several reasons.

- We believe we are the first group to integrate these multiple dimensions in drosophila.
- We have demonstrated that this approach is sound.
- We have used the network to help guide the discovery process of characterizing two unknown genes.
- We have discovered that each data set, *e.g.* Arbeitman’s time-course, tends to naturally aggregate with itself. To integrate the data, we queried for paths that included a mix of datasets, explicitly ignoring paths that were only drawn from one data set.

The remaining paper is as follows. In Section 2 we introduce and provide a brief description of the data sets used. In Section 3 we discuss issues of combining systems data. Next in Section 4 we provide some insight into how inferences are made from the network. In Section 5 we present how we verified our network. In Section 8 we present how we came about elucidating two currently, unannotated genes. In Section 7 we summarize our work and discuss our future work.

2 Datasets and Methods

Every genetic or genomic experimental technique offers the biological researcher a different perspective into how genes interact on different levels, or under different experimental conditions. The context of the experiment offers results that are often very informative to the question that was posed, but can also offer us insight into other experimental data.

The data sets used in this application are microarray co-expression assays (section 2.1), protein-protein interactions (section 2.2), genetic interactions (section 2.3), human curated phenotypic data (section 2.4), and transcription factor binding sites (section 2.5).

2.1 Gene Co-Expression

Microarray expression data quantifies the relative steady state mRNA level of a large proportion of the transcriptome. Each of the data sets used here systematically quantified gene expression across several biological states: time course of development, dissected organs, and two sexes.

The gene network currently contains three microarray experiments that were taken from Gene Expression Omnibus (GEO) ¹; Life cycle of *Drosophila melanogaster* experiment: M. Arbeitman, *et al.*, Paucity of Genes on the Drosophila X Chromosome Showing Male-Biased Expression: M. Parisi, *et al.* [17], Tissue-specific gene expression and ecdysone-regulated genomic networks in drosophila: T. Li, *et al.* [13]

Each of the microarray data sets were taken in their raw data form, then internally normalized to address noise effects inherent to microarray experimentation. Normalization was done using the OLIN [7] package in the

¹<http://www.ncbi.nlm.nih.gov/geo/>

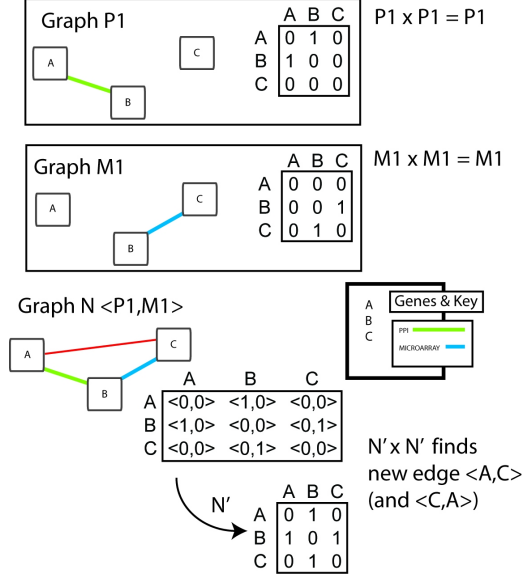


Figure 1: In this stylized example we demonstrate how our system brings together data and produces new edges. The non-integrated data sets are P1 and M1. We are interested, say, in establishing whether there is evidence for some relationship between A and C. We depict the relationships using Boolean matrices where 0 indicates no relationship and 1 indicates there is. Observe that producing the two matrices does not produce new edges (modulo 2 arithmetic). By combining the two data sets into vectors and some operator, say, $\max(x,y)$, the vector value will be 1 if there is a 1, and 0 otherwise. Now producing the new matrix will produce a new edge from A to C adding paths that are supported by connected evidence.

R² statistical programming environment. Next, the FOCI algorithm, (a complete description of the algorithm can be found at [15]), was applied to each individual microarray study to find correlation values for a gene-gene relationship in that particular study. Each microarray experiment was taken separately since each study was done under different experimental conditions and technical variation and cross experiment combination of microarray data has been found to be unreliable[26, 10].

2.2 Protein-Protein Interaction

Protein-protein interaction data provides some of the strongest evidence for making a functional connection between genes since the actual gene products are physically interacting.

The PPI data set was compiled into non-redundant set of strictly protein-protein interactions from the four major protein interaction databases: FlyGrid³, DIP⁴, MINT⁵, and BIND⁶. The experimental evidence for protein-protein interactions were primarily yeast-2-hybrid assays with a handful of pull down experiments. The most numerous contribution to the data set was from the Giot, *et al.* [12] yeast-2-hybrid proteome screen. However, due to the high false positive rate inherent to system, these data were given a lower weight in establishing network edges.

2.3 Genetic Interaction

Genetic interaction experiments are widely regarded by geneticists as a reliable method for discovering functional relationships between genes. The genetic interaction data was obtained from Flybase⁷, where experts curate data from the primary literature and assign genetic interactions to one of two main classes: a relationship between two

²<http://www.r-project.org/>

³http://biodata.mshri.on.ca/fly_grid/servlet/SearchPage

⁴<http://dip.doe-mbi.ucla.edu/dip/Download.cgi?SM=3>

⁵<http://mint.bio.uniroma2.it/mint/>

⁶<http://ftp.blueprint.org/pub/BIND/data/divisions/asn.binary>

⁷<http://flybase.bio.indiana.edu/>

genes that is either “tested and found” or “tested and not found.” While the data in Flybase is curated on the level of alleles, we compiled these to the level of the gene.

Genetic interaction data indicates that, on some level, a gene has a functional connection to another gene, though we might not be sure of what “level,” *e.g.*, a transcriptional, post-transcriptional, translational, post-translational, or metabolite level.

The genetic interaction data in the network consist of approximately 5.2K “tested and found” relationships and 1.3K “tested and not found” relationships. Lastly, since there is a great deal of confidence in genetic interaction experiments, a functional relationship can be confidently drawn as an edge in the network between two genes that share a genetic interaction.

2.4 Allelic Phenotype

Phenotypic data includes a description of the cellular or organismal (or both) defects that have been observed for mutant alleles of a particular gene. The phenotypic data in Flybase is drawn from the primary literature by expert curators and is captured as both controlled text annotation of phenotypic class and affected body parts and free text descriptions [6].

We attempted to quantify the similarity between genes based on the controlled text phenotypic information using cosine similarity measures [4]. In order to gauge the informativeness of the cosine similarity measures we tested the similarity between phenotypic correlation values and GO terms and ultimately found it to be a poor correlation. Consequently we did not use this data source. Instead, the phenotypic data was used to *strengthen* edges that are established on the basis of other data sources. This is the same approach used for the TFBS data (see section 2.5).

2.5 Transcription Factor Binding Site

In assessing functional relationships between genes, transcription factor binding sites (TFBS) can be used to augment the existing network. It is known that co-regulated genes share a significant number of regulatory elements [14, 24]. Yet, because identifying TFBS from the annotated *Drosophila* genome sequence is computational, and because of the limited number of experimental TFBS data sets, we found that this information is best used as supporting evidence, rather than used as evidence for creating new edges between genes.

TFBS data for this project were extracted from flyreg.org⁸, where Casey Bergman *et. al.* [3] mined the literature related to DNase I footprints in *Drosophila*. Seventy-five position weighted matrices (PWM) were created from the motifs found in the promoter regions of genes studied by Bergman. The PWM search program Patser⁹ was then run on all promoter regions from the version 4.1 annotation of the *Drosophila* genome. Correlation values were finally calculated for all pairs of genes to determine their number of shared promoters.

3 Integration

Systematic (genome-wide or approaching genome-wide) functional genomics data can be used to infer gene function. The data set can be analyzed to group together (cluster) genes that behave similarly under the experimental conditions. The clustering of previously un-annotated genes together with genes of known biological function can be used to posit hypotheses regarding the biological functions of those un-annotated genes. This has been described as guilt by association. This has been demonstrated to be case for microarray expression data [1, 13, 17], protein-protein interaction data [12], transcription factor binding sites [24], phenotypic data [6] and genetic interactions [5]. In some of these data sets there are molecular models implicit in the functional connections drawn inferred between genes. For instance the presence of a TFBS suggests transcriptional regulation, and protein-protein interaction screens suggest the two genes interact via direct protein-protein contact. On the other hand functional relationships inferred on the basis of other experimental data have no implicit molecular models. For instance similar phenotypes or genetic interactions can be found between genes that have a variety of underlying molecular connections. Thus it useful to define the term functional connection as describing a relationship between two genes that are in a

⁸<http://www.flyreg.org/>

⁹http://rsat.ulb.ac.be/rsat/patser_form.cgi

common genetic pathway (analogous to the genetic interactions) and/or participate in a common biological process (analogous to biological process as described by GO consortium). Hereafter we use this definition.

Combining different sources of functional genomics data can be used to both mutually support the inferences made from individual data sets as well as to further refine the grouping of genes into functionally coherent clusters. This has been demonstrated by combining data such as microarray expression profiles, protein-protein interaction data, phenotypic data and genetic interaction data in yeast and *C. elegans* [9, 19, 25]. The rationale for combining the data includes the following. First, each of the data sources has inherent shortcomings including false positives and false negatives. To the extent that different experimental approaches have different biases in false positives and negatives then combining data has the effect of canceling noise and summing the signal. Second, for purely practical reasons the various data sources have different sets of missing data. Third, combining data leads to the grouping of smaller, and probably more coherent, clusters. Fourth, some experimental approaches are only capable of discerning a subset of all possible molecular connections. For instance PPI can only detect direct interactions between proteins and microarray data is only capable of detecting changes in steady state RNA levels. Thus paths with alternating molecular connections may be missed in single experimental approaches. For instance, if gene *A* has a protein-protein interaction with gene *B*, which in turn regulates the transcription of gene *C*, then the indirect functional connection between gene *A* and *C* may not be detected in either PPI data or microarray data. In summary, combining the data should increase the confidence of functional connections, discover connections that might not be detected by individual experimental approaches, and refine clusters to smaller more cohesive sub-clusters.

Given that some of the data sources are global and some are conditional we are faced with the choice of building networks of all possible functional connections, or confining networks to those detected under specific conditions (such tissue or cell type, disease state, developmental state). Both approaches have merit. As a first pass we have elected to build a global network of all possible connections since this will be informative to investigators studying different systems.

4 Data Inferencing

Our data model is designed around a number of important aspects. First, the integration of data must be biologically sound as discussed in the previous section. Second, the model needs to be extensible. There are many new sets of data, we imagine, that will be produced by biologist in the future. We want to ensure this new data can be transparently and easily incorporated into the existing database and network. The data model needs to be able to handle these data. As mentioned in the introduction of the paper, we opted for a simpler initial, extensional design. This decision was made from the standpoint of easy integration, as mentioned here, but also because it places fewer explicit requirements on us as we bring together the data. Furthermore, one of the more useful inference procedures (logic-theoretic *modus ponens*) can be implemented relatively easily, partially in database Standard Query Language (SQL).

Essentially, we are performing a kind of deduction. Suppose we have three genes *A*, *B*, *C*, and we have evidence of interactions $\langle A, B \rangle$ and $\langle B, C \rangle$. Then we can conclude (logically) that there is an interaction $\langle A, C \rangle$. The “degree” of our belief (or what the evidence suggests) can be adjusted depending on many kinds of logics that are available—ranging from classical 2-valued (Boolean) to multi-valued logics where discrete truth values are replaced with values over the interval [0,1]. Probably the most widely known of these is the so-called “fuzzy” logic of Zadeh [27]. Through this procedure we can add new edges directly altering the graph—new paths are formed. While a complete detailed discussion of our inferencing procedure is not possible here, we will describe some key elements. Most of the simple deductive procedures (without negation) usually are a form of transitive closure. More formally, given a graph $G = \langle V, E \rangle$, the transitive closure, denoted G^* is $G^* = G \cup G^2 \cup G^3 \cup \dots \cup G^{|V|-1}$ where each product adds a set of new edges. Because the graph is finite, there can only be a finite number of edges. In Figure 1 we present a stylized version of our approach assuming there are three genes *A*, *B*, *C*, where as presented in the previous section, a relationship appears from the integration of the data. More formally, each edge is a vector of interactions $\langle \mu_1, \mu_2, \mu_3, ppi, gi, anp, tfbs, rnai \rangle$, where each interaction is drawn from a database where the interaction is stored and analyzed. There are three microarray assay values $\mu_{1,2,3}$, protein-protein *ppi*, genetic interaction *gi*, annotated phenotype *anp*, transcription factor binding site *tfbs*, and RNAi *rnai* (although we have not completed incorporating this dimension at this time).

5 Verifying Gene Network Topology

In studying gene networks one must consider both its strengths *and* its weaknesses. Probably most importantly, we must demonstrate that by integrating the different data sets, we are adding *sufficient* value to the predicted functional connections between two genes, or sets of genes.

The assumptions made and inferences that can be drawn from individual data sets are briefly touched on in Section 4.

Knowing that each data set has strengths and weaknesses, we can leverage each data set against each other to create a much more informative data set. Gunsalus, *et al.* [9] used early embryogenesis of *C. elegans* to show that integrating microarray, protein-protein interaction, and RNAi data brings out relationships of “molecular machines” between genes that could not be identified through individual data sources otherwise. Kemmeren, *et al.* [11] used multiple high-throughput *S. cerevisiae* data sets to show that by combining the data sets, one could reasonably predict gene function of unknown genes. To date, the authors are not aware of any studies of this comprehensive nature in *Drosophila*, so by building a gene network from *D. melanogaster* specific data allows the expansion of knowledge not only within the *Drosophila* research community, but also allows comparison of prevalent “molecular machines” across species.

Given the research that has been done on gene networks in *C. elegans* and *S. cerevisiae*, it is apparent that combining disparate experimental biological data sets can lead to some interesting insight into the functioning of genes. It must now be shown that our network created from *Drosophila* data is being reasonably created. However, this type of validation is not simple, as there is no baseline genome-wide network that exists to compare against, which will give us a reasonable cross-species measure of significance. Also, due to the enormous size of the datasets, the great deal of already intractable problems become even harder. Above all, biological phenomenon do not follow parametrized rules that can be easily made true for every instance of an occurrence, so meaningful quantifications of these phenomena are hard to come by.

The functional relatedness of any two genes taken from any data source being integrated into the network can have biological reasons for not being related. For example, if proteins from two genes are shown to have a positive interaction in a yeast two hybrid study, they may never come in contact with each other due to developmental or localization effects in the cell. Thus, the positive interaction from the experimental data does not reflect the actual biology. Another example can be illustrated with microarray data, where if there is high correlation between two genes from microarray data, we can not say that these two genes are necessarily functionally related, because those two genes may be transcribed in great abundance at the same time, but for completely orthogonal functions(i.e. one is involved in pathway x and the other is involved in pathway y).

Knowing the problem of validating the gene network, we highlight 2 measures that were implemented to verify that the gene network is providing new and insightful information that could not be seen through any individual data set. First, we compare the data taken from high-throughput data sets to the gene-gene relationships of genetic interactions(section 5.1). Secondly, we use GO terms to identify similar functionally annotated genes and test connectivity of these genes (section 5.2).

5.1 Detecting Genetic Interactions From Other Data Sources

The features that make genetic interaction data a standard of genetic research (see section 2.3) also allow us to use this data set as a baseline for verifying network connectivity. Although genetic interaction experiments are specific to a very small subset of genes, when we combine all the genetic interaction data, we have a set of over 6,500 gene-gene relationships to work with.

Genetic interactions come in two forms, “tested and found” or “tested and not found”, which can be treated in a sense like true positives and true negatives. Although there are biological subtleties that do not completely support treating genetic interaction data as a baseline, it is the most well informed set of data we have.

Table 5.1 illustrates the progression of connections to the genetic interaction data that can be recovered by individual data sets, across data sets, and using an $A-x-B$ connection. As shown in figure 1, The $A-x-B$ connection is any connection of path length 2, where A and B are the nodes of interest and x is any node that connects A to B , where $x \neq A$ or $x \neq B$.

It can be seen that as individual data sets, the Parisi microarray data show the most number of connections, while the Arbeitman microarray data set has the least. When the union of all the data sets, noted $\mathcal{A}$ is taken, we can detect 37% of the “tested and found” genetic interactions. When the $A-x-B$ relationship is queried, we can

Category	Count GI tested and found	Count GI tested not found	% tested found	% tested not found
All GIs in data	5225	1292	100	100
Arbeitman	78	24	1.5	1.8
Larval	857	224	16.4	11.6
Parisi	1375	376	26.3	29.0
PPI	130	23	2.5	1.8
$\mathcal{A} = \text{Arb} \vee \text{Par} \vee \text{Lar} \vee \text{PPI}$	1935	499	37.0	38.6
Path Length 2	3374	867	64.6	67.1
Path Len 2 $\vee \mathcal{A}$	3401	874	65	67.6

Table 1: Illustrates the relative contribution of individual data sets as compared to different methods of combining the data with respect to retrieving known genetic interactions from the network. It can be seen that by integrating data sources, the number of genetic interactions retrieved is increased and when looking for $A-x-B$ relationships, or path length 2, the percentage of genetic interactions increase. Each individual data set and the combination of data sets increase the “tested and found” as well as “tested and not found” genetic interactions.

retrieve 64.6% of the “tested and found” relationships, while the union of $\mathcal{A}$ and path length 2 slightly increased our retrieval rate to 65% of the total “tested and found” genetic interactions.

Table 5.1 illustrates an important and subtle point about data integration; namely, that when a single or even multiple experiments are considered, overlap between those experiments (in terms of the number of functional connections drawn) is low, at most about 26%. However, when considering what we call path length 2 connections (*e.g.*, gene A to B are connected by a protein-protein interaction, and B and C by microarray coexpression), the overlap between experiments is much higher, around 65%. That is, if we accept the biological validity of the connections between A and B and between B and C as being in some sense equivalent, our approximation of ground truth in this example is more than twice as good. One key point to notice from table 5.1 is that the percentage of “tested and found” relationships that are retrieved through each data set or integrated data sets are almost identical to the percentage of retrieved “tested and not found” relationships. There are several biological or procedural reasons that can explain this phenomenon.

From a biological perspective, one can see that the majority of our functional relationships are taken from microarray co-expression experiments. Microarray experiments tell us what genes are being co-expressed, but this does not definitively mean that the genes are functionally connected, as there are temporal and spacial issues to consider. Also genes may simply be involved in different pathways, but happen to be expressed at the same time. So, inferencing a connection between two genes from microarray data may not hold true with respect to genetic interaction data. However, this is not negating the microarray data when making functional connections, as there could be some other reason that the microarray data shows a relationship that a genetic interaction may not (*i.e.* any experimental condition that that is not similar to the conditions the genetic interaction was tested under).

From a procedural perspective, one can see that our data set consists of 5,200 “tested and found” and 1,300 genetic interactions, so there are exactly 4 times as many “tested and found” relationships than “tested and not found.” This discrepancy in numbers can be attributed to many factors, such as “tested and not found” results are not published as often, but the difference in numbers can attribute to the trends that are observed from our data set. Lastly, a “tested and not found” genetic interaction result is not infallible, as a phenotype may have occurred from a genetic interaction between two genes, but may not be too subtle to observe by the research.

5.2 Network Clustering Using Biological Process

The Gene Ontology (GO) [8] identifies a controlled vocabulary to annotate genes in three class; Biological Process, Cellular Component, and Molecular Function, where Biological Process is the class of interest in this assessment. Presumably, experimental evidence would support the classification of a certain gene with a particular GO term, therefore a group of genes annotated with the same GO term should have experimental evidence of this annotation, albeit the annotation of a gene may be supported by disparate data sources.

Knowing that genes annotated with the same GO term are likely to share similar experimental evidence, a group of genes in the gene network that share a common GO term should show significantly increased connectivity

	Arbeitman	Parisi	Li	PPI	Gen. Inter.
Percentage	6.5%	21.5%	8.6%	10.4%	16.3%
Std. Dev.	4.7%	10.0%	7.6%	9.0%	20.5%

Table 2: Table showing how connected a set of genes are that share a common GO term along with how variable each data set can be. The values shown are the average percentage of nodes that are interconnected among 85 GO clusters (see section 5.2) as related to each specific dataset.

over a group of randomly selected genes from the entire data set, where connectivity can be simply measured by the percentage of genes that share an edge in a given set.

Each gene in the data set was queried against FlyBase to retrieve its annotated Biological Process GO term(s), if they are present. From all the annotated genes found in the data set, there were 85 groups with 50 to 300 genes that shared a single GO term. Within each GO term cluster of genes, direct connections between the genes were identified and the percentage of genes being interconnected is used as the measure of how strong a GO term cluster is connected. For example, in the data set there were 175 genes that are annotated with *GO:0007399; Nervous system development*, where 130 genes were found to be connect, thus being 74% connected. Measures were taken for all 85 clusters and on average, where the clusters had 61.6% of the genes being connected with a standard deviation of 18.3%.

In order to demonstrate the GO term clusters are, in fact, enriched for connected genes, a comparison to several random models was done. Clusters of 100, 200, and 300 genes were selected at random from the entire data set and then tested for the percentage of genes that are connected. This test was run 100 times for each of the 3 sets of genes, where new random sets were generated on each run. For the group of 100 random genes, on average, 26.6% of the genes were connected with standard deviation of 5.4%. For the group of 200 random genes, on average, 30.5% of the genes were connected with standard deviation of 3.8%. For the group of 300 random genes, on average, 30.9% of the genes were connected with standard deviation of 3.4%.

An illustration of a randomly generated network as compared to a GO term cluster can be seen in figure 6. The images were created using Cytoscape [21] and the network representing a GO term cluster was generated from 275 genes that share the GO term *GO:0006468; Protein amino acid phosphorylation*.

The connectivity measures were normalized for each GO term cluster, as well as the random sets, in order to directly compare if the GO term clusters are significantly more connected than expected from the random model. Figure 6 shows this comparison of each of the 85 GO term clusters through z-score, where the z-score is calculated as $z_i \equiv \frac{x_i - \bar{x}}{\sigma}$. If a z-score of 2 is considered the cutoff value for significance with respect to the GO term clusters having higher connectivity than randomly expected, then for all three of the random model comparisons, 91% of the GO term clusters were significantly higher than expected randomly.

The last step in demonstrating that the integration of data does, in fact, increase our knowledge base in the form of a network is to show that individual data sets on their own can not create the number of connections that the network can with respect to the GO term clusters. Looking at table 5.2, one can see that the measure of connectivity found in any given data set is considerably lower than the network as a whole, which is roughly 64% connected. It can also be seen that each data set has a high degree of variability in making connections within specific GO term clusters, which probably reflects the conditions of the experimental data.

This method of using groups of genes segmented by GO Biological Process terms shows that the combining of the data strengthens the argument that integrating the data allows for increased biological insight.

6 Discovery

An example of bringing together disparate datasets into a network that can be queried for biological discovery is shown in Figure 8. After constructing a global gene network incorporating microarray, PPI, GI, Phenotypic and TFBS data we selected genes annotated for a particular category of the gene ontology (GO term), in this case with *GO:0006281; DNA repair*. We then used a simple set of logical rules to query the database for other non-annotated genes that were also highly connected to the members of this sub-network. The result is a sub-network of genes joined by all of our data sources: microarrays, protein-protein interactions, genetic interactions, transcription factor

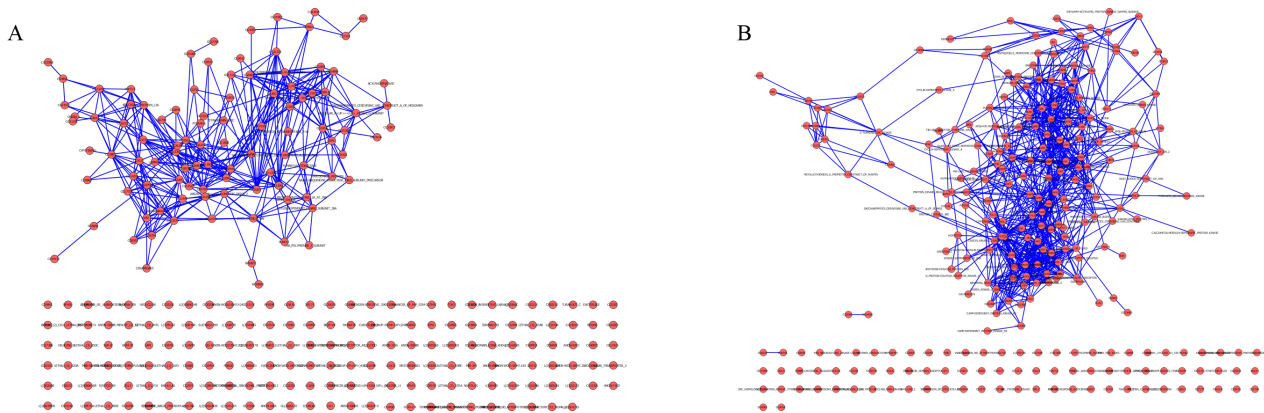


Figure 2: *A* is a network created with 275 randomly selected genes from the entire data set, while *B* is a network created from the 275 genes annotated with the GO term *GO:0006468; Protein amino acid phosphorylation*. The random network shows that 37% of the 275 genes are connected, where 69% of the 275 genes in the GO term annotated cluster are connected. 91 functionally related genes are more tightly clustered in the gene network than at random.

binding site (TFBS) similarity, and phenotype similarity. Two genes of particular interest- *CG6171* and *CG14781*- were found to be new additions to the network, and are sparsely annotated regarding putative function. Our tentative hypothesis is that these two genes play a role in DNA repair pathways.

Examining *CG6171* and *CG14781* for nearest-neighbors using only high-confidence data (genetic interaction, protein-protein interaction, or microarrays correlated above a correlation of 0.95) reveals connections to proteins with annotations other than DNA repair. It is not clear using this limited type of query whether DNA repair is a reasonable functional imputation. However, when combined with the GO-annotated network, these genes show high correlations to multiple DNA repair proteins, often with two or three datasets. For example, *CG6171* has microarray correlations of 0.96 from two different experiments to the FLAP endonuclease, and microarray correlations of 0.90 with *Mat1*, *Mei9*, *CG12018*, and *cryptochrome*, all annotated as DNA repair proteins. In FlyBase, the only explicit evidence regarding *CG6171* is a protein-protein interaction with *Traf3*, a defense-response protein. By itself, this is not compelling; in light of the known roles of other proteins highly correlated in expression with *CG6171*, such as response to UV-induced DNA damage, the observation takes on added meaning.

In looking more closely at *CG6171*, we found that *C. elegans* and *S. cerevisiae* lack any close homologs to the *D. melanogaster* protein (e-values of 0.06 and 0.2, respectively). In contrast, both *H. sapiens* and *M. musculus* have much more convincing homologs, at $1e^{-14}$ and $5e^{-19}$, respectively. It is interesting and noteworthy that these human and mouse orthologs have no functional information associated with them. Thus, our functional imputation of DNA repair is the first and only experimental evidence ever provided. In addition, using worm or yeast datasets for this purpose would not be fruitful due to absence of an identifiable counterpart in these organisms.

In searching for similar homology information about *CG16781*, we found that among *A. gambiae*, *C. elegans*, *S. cerevisiae*, *M. musculus*, and *H. sapiens*, there are roughly 100 proteins in each genome showing some homology ($1e^{-05}$ to $1e^{-14}$). Within this large number of potentially similar proteins, a wide variety of biological functions are represented. It is not obvious from sequence similarity alone which if any of these possibilities is more likely. Only by using experimental data from multiple sources do we discover an intimate link with DNA repair proteins.

The simple example in Figure 8 illustrates several important principles of the Indigene project. First, it is clear that, given data from FlyBase and the NR database about close homologs to *CG6171* and *CG16781*, a reliable prediction about the function of these proteins would not be possible. However, by combining data from multiple sources, using simple logical queries designed by the biologist to return specified subsets of data, and employing domain knowledge and intuition, we have arrived at testable predictions for the function of two hitherto unknown proteins. Furthermore, the predictions depend on both the data and phylogenetic position of *D. melanogaster*. Most importantly, given the critical importance of DNA repair to numerous human disease pathways such as cancer, we have illustrated a logical and compelling way to generate new biological insight using experimental data from the model organism *D. melanogaster*.

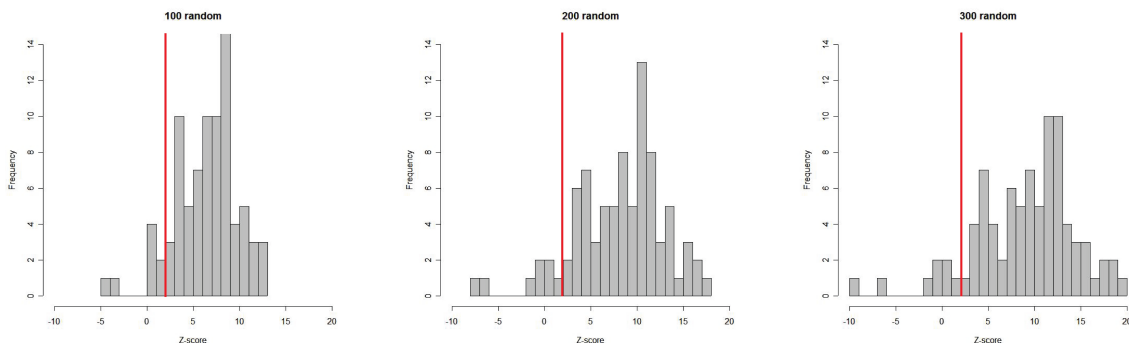


Figure 3: These bar charts represent the significance scores of the 85 GO term clusters of genes as compared to randomly selected sets of genes pulled from the entire data set, where the abscissa is the z-score and the ordinate is the frequency distribution. Significance for the clusters is measured by the percentage of genes in the set that are interconnected. Each of the charts show that 91% of the GO term clustered genes are more significantly connected than compared to the random model, which is indicated by anything to the right of the red line, which marks a z-score of 2. The leftmost chart represents the 85 clusters compared to a random set of 100 genes. The middle chart represents the 85 clusters compared to a random set of 200 genes. The rightmost chart represents the 85 clusters compared to a random set of 300 genes.

7 Summary

The ultimate elucidation of the functions of all genes in a metazoan and the interactions between them in cell types over time is going to involve drawing on multiple orthogonal data sources. This will also rely heavily on comparative approaches across taxa [22]. Methods of analysis need to be flexible both to incorporate new, and as yet unanticipated experimental data, and also to allow the extant data to be queried and navigated by Biologists with expert domain knowledge. It will be essential that this be pursued in genetic model organisms so that the imputed hypothesis can be tested directly for further refinement to network models.

Here we have presented Indigene a conceptually simple yet demonstrably effective method for integrating disparate types of genome wide functional data. It is highly adaptable to accommodate new sources of experimental data as they continue to become available. It is also flexible in allowing different weights to be applied to the various data sets. This will allow parameter settings to be optimized for particular conditions. In the very simplest application this integration provides a simple navigable summary of the various experimental data supporting functional connections between all pairwise combinations of genes. This is valuable tool allowing users to quickly visualize the previously identified connections between genes, which is increasingly necessary as genome wide experiments draw domain specialists into new and unfamiliar territories. Beyond the simplest function of data exploration, we have demonstrated that combining the data can lead to discoveries not possible without integration. This is evidenced by (i) increased ability to retrieving known, (ii) enriched interconnections within sub-networks of genes sharing GO annotations, and (iii) a specific example of functional annotation imputed on the basis of a high degree of interconnection with a sub-network of genes with previously known biological functions.

We have applied Indigene to integrate functional data in the model organism *Drosophila melanogaster*. To our knowledge this is the first such study in this organism. Given that flies are among the preeminent model metazoan with tractable forward and reverse genetics it is undoubted that flies will contribute significantly to the systems approach of gene network exploration. While the fly data is certainly sparser than that for yeast in which most systems approaches are being pursued, it has the merit of a complex metazoan. We anticipate that ongoing functional studies in flies will allow us to further refine this first pass at a global gene network.

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References

- [1] M. Arbeitman, E. Furlong, F. Imam, E. Johnson, B. Null, B. Baker, M. Krasnow, M. Scott, R. Davis, and K. White. Gene expression during the life cycle of drosophila melanogaster. *Science*, 297(5590):2270–2275, 2002.
- [2] J. Bader, A. Chaudhuri, J. Rothberg, and J. Chant. Gaining confidence in high-throughput protein interaction networks. *Nat. Biotechnol*, 22:78–851, Dec. 14 2003.
- [3] C. Bergman, J. Carlson, and S. Celniker. Drosophila dnase i footprint database: A systematic genome annotation of transcription factor binding sites in the fruitfly, d. melanogaster. *Bioinformatics*, 21:1747–1749, 2005.
- [4] M. W. Berry. *Survey of Text Mining: Clustering, Classification, and Retrieval*. Springer-Verlag New York, inc., New York, NY, 2004.
- [5] T. F. Consortium. The flybase database of the drosophila genome projects and community literature. *Nucleic Acids Research*, 31:172–175, 2003.
- [6] R. Drysdale. Phenotypic data in flybase. *Briefings in Bioinformatics*, 2:68–80, 2001.
- [7] M. Futschik and T. Crompton. Olin: optimized normalization, visualization and quality testing of two-channel microarray data. *Bioinformatics*, 21(8):1724–1726, 2005.
- [8] Gene Ontology Consortium. Creating the gene ontology resource: design and implementation. *Genome Research*, 11(8):1425–1433, 2001.
- [9] K. C. Gunsalus, H. Ge, A. Schetter, D. S. Goldberg, J. Han, T. Hao, G. Berriz, N. Bertin, J. Huang, L. Chuang, N. Li, R. Mani, A. Hyman, B. Sonnichsen, C. Echeverri, F. Roth, M. Vidal, and F. Piano. Predictive models of molecular machines involved in caenorhabditi elegans early embryogenesis. *Nature Letters*, 436:861–865, August 11 2005.
- [10] A. Jarvinen, S. Hautaniemi, H. Edgren, P. Auvinen, J. Saarela, P. Kallioniemi, and O. Monni. Are data from different gene expression microarray platforms comparable? *Genomics*, 83:1164–1168, 2004.
- [11] P. Kemmeren, T. Kockelkorn, T. Bijma, R. Donders, and F. Holstege. Predicting gene function through systematic analysis and quality assessment of high-throughput data. *Bioinformatics*, 21(8):1644–1652, 2005.
- [12] e. a. L. Giot. A protein interaction map of drosophila melanogaster. *Science*, 302(5651):1727–1736, 2003.
- [13] T. Li and K. White. Tissue-specific gene expression and ecdysone-regulated genomic networks in drosophila. *Developmental Cell*, 5(1):59–72, July 2003.
- [14] Z. Lubovac and B. Olsson. Towards engineering of genetic regulatory networks. Technical report, Department of Computer Science, University of Skovde, 2003.
- [15] P. Magwene and J. Kim. Estimating genomic coexpression networks using first-order conditional independence. *Genome Biology*, 5(12):R100, 2004.
- [16] E. Marcotte, M. Pellegrini, M. Thompson, T. Yeates, and D. Eisenberg. Detecting protein function and protein-protein interactions from genome sequences. *Science*, 285(5428):751–753, July 30 1999.
- [17] M. Parisi, R. Nuttall, D. Naiman, G. Bouffard, J. Malley, J. Andrews, S. Eastman, and B. Oliver. Paucity of genes on the drosophila x chromosome showing male-biased expression. *Science*, 299(5607):697–700, 2003.
- [18] J. Pearl. *Probabilistic Reasoning in Intelligent Systems: Networks of Plausible Inference (Revised 2nd Ed.)*. Morgan Kaufmann, San Francisco, CA, 1988.

- [19] F. Piano, A. Schetter, D. Morton, K. Gunsalus, V. Reinke, S. Kim, and K. Kemphues. Gene clustering based on rnai phenotypes of ovary-enriched genes in *c. elegans*. *Curr. Biol.*, 12(22)(22):1952–1958, November 19 2002.
- [20] B. Schwikowski, P. Uetz, and S. Fields. A network of protein-protein in yeast. *Nat. Biotechnol.*, 18(12):1257–61, Dec. 18 2000.
- [21] P. Shannon, A. Markiel, O. Ozier, N. Baliga, J. Wang, D. Ramage, N. Amin, B. Schwikowski, and T. Ideker. Cytoscape: A software environment for integrated models of biomolecular interaction networks. *Genome Research*, 13:2498–2504, 2003.
- [22] J. Stuart, E. Segal, D. Koller, and S. Kim. A gene-coexpression network for global discovery of conserved genetic modules. *Science*, 302(5643):249–255, Oct 10 2003.
- [23] O. Troyanskaya, K. Dolinski, A. Owen, R. Altman, and D. Botstein. A bayesian framework for combining heterogeneous data sources for gene function prediction (in *s. cerevisiae*). *PNAS*, 100(14):8348–8353, July 8 2003.
- [24] J. van Helden, D. Gilbert, L. Wrnisch, M. Schroeder, and S. Wodak. Application of regulatory analysis and metabolic network analysis to the interpretation of gene experection data. *Lecture Notes in Computer Science*, 2066:155–172, 2000.
- [25] A. Walhout, J. Reboul, O. Shtanko, N. Bertin, P. Vaglio, H. Ge, H. Lee, L. Doucette-Stamm, K. Gunsalus, A. Schetter, D. M. K. Kemphues, V. Reinke, S. Kim, F. Piano, and M. Vidal. Integrating interactome, phenotype, and transcriptome mapping data for the *c. elegans* germline. *Curr. Biol.*, 12(22):1959–1964, November 19 2002.
- [26] H. Wang, X. He, M. Brand, C. Wilson, and L. Liu. A study of inter-lab and inter-platform agreement of dna microarray data. *BMC Genomics*, 6(71), 2005.
- [27] L. Zadeh. Fuzzy sets. *Information and Control*, 8:338–353, 1965.

Appendix A

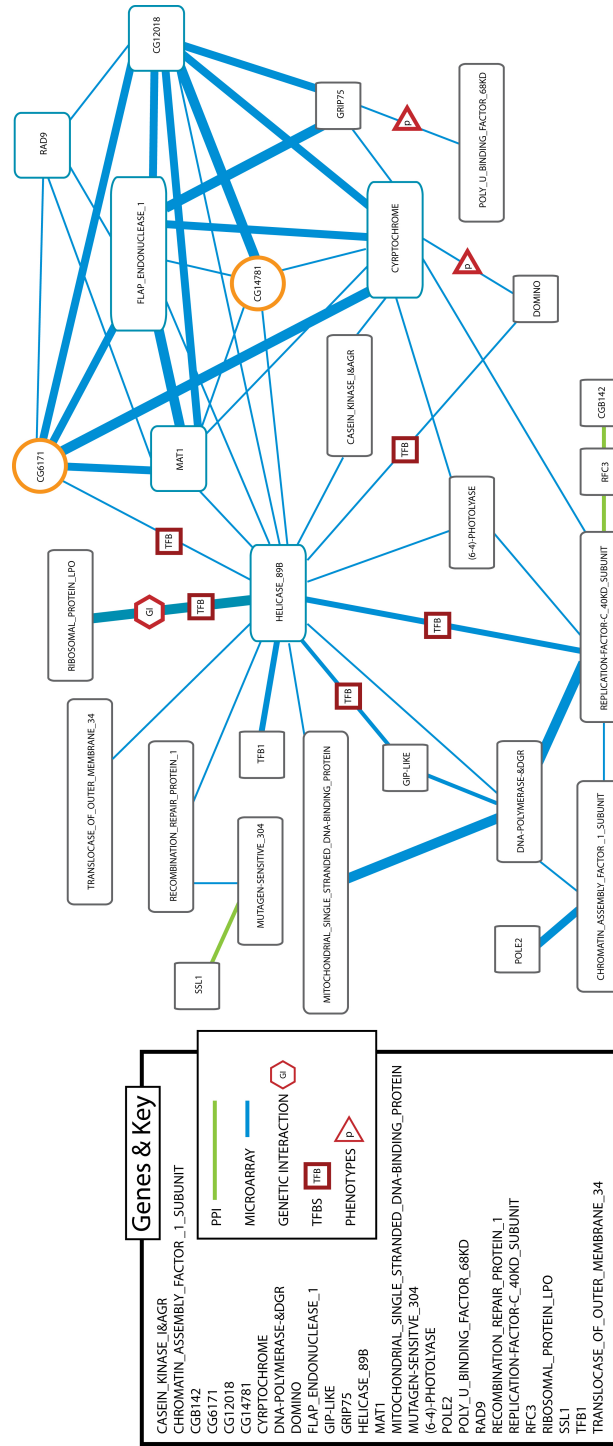


Figure 4: This network shows how two unknown genes, depicted as yellow circles, can be brought into context through the network showing high connectivity to characterized genes. Edges represent kinds of relationships: blue for microarray and green for PPI. The thickness of the line approximates the strength of the connection. The lines have been further decorated with a square if they share TFBS, hexagons for genetic interaction, and triangle for similar phenotypic patterns. We depicted the immediate neighbors of the unannotated genes with blue boxes.